

Project Name: IoT Based Fish Pond Monitoring System (Structured Report)

1. Introduction

- IoT based system for monitoring fish pond water quality
- Designed for real-time data collection and remote monitoring
- Targets common problems in Bangladesh aquaculture
 - sudden fish death
 - poor water quality
 - lack of continuous monitoring

2. Objectives

- Monitor water parameters continuously
- Reduce fish mortality
- Enable remote monitoring via WiFi
- Provide alert system for dangerous conditions
- Allow future automation (aerator, feeder)

3. System Working Principle

- Sensors collect water data
- ESP32 processes the data
- Data sent via WiFi to cloud
- User views data on mobile/web
- Alerts generated if values exceed safe range

4. Main Hardware Components:

4.1 ESP32 Development Board

- Role: Main controller and WiFi communication
- Features
 - Built-in WiFi and Bluetooth
 - Dual core processor
 - Low power consumption
- Price: 430

Reference: <https://store.roboticsbd.com/development-boards/948-esp32-esp-32s-30p-nodemcu-development-board-wireless-wifi-robotics-bangladesh.html>

Alternative

- NodeMCU ESP8266
 - Limitation: Less powerful
 - Price: 350
 - Reference: <https://store.roboticsbd.com/development-boards/694-esp8266-ch340-nodemcu-wifi-module-lua-v3-robotics-bangladesh.html>

4.2 Temperature Sensor (DS18B20 Waterproof)

- Role: Measures water temperature
- Features
 - Waterproof probe
 - High accuracy $\pm 0.5^{\circ}\text{C}$
 - Wide range
- Price: 220
- Reference: <https://www.daraz.com.bd/products/ds18b20-waterproof-temperature-sensor-i117163000.html>

4.3 pH Sensor (PH 4502C)

- Role: Measures acidity or alkalinity
- Features
 - Range: 0 to 14 pH
 - Analog output
 - Needs calibration
- Price: 2900
- Reference: <https://techshopbd.com/product/ph-sensor-with-module>

Alternative

- Gravity pH sensor
 - Price: 3750
 - More stable
 - Reference: <https://www.bdtronics.com/gravity-analog-ph-sensor-meter-kit-v2-compatible-with-3-3v-and-5v-main-controller-for-water-quality-testing-aquaculture.html>

4.4 Turbidity Sensor

- Role: Measures water clarity
- Features
 - Detects suspended particles
 - Indicates pollution or algae
 - Price: 820 bdt
 - Reference: <https://techshopbd.com/product/water-turbidity-sensor-for-arduino>

Alternative

- turbidity sensor
 - Price: 965
 - Better accuracy
 - Reference: <https://robodocbd.com/product/turbidity-sensor-module>

4.5 Relay Module

- Role: Controls external devices
- Features
 - Switch high voltage devices
- Module Names:
 - Aerator Or
 - Water pump
- Price: 400
- Reference: <https://shorturl.at/NnxJU>

Alternative

- Solid state relay
 - Faster switching
 - Slightly higher cost

4.6 Buzzer and LED

- Role: Local alert system
- Features
 - Low power
 - Simple digital output
- Price: 200

- References: <https://store.roboticsbd.com/tools-accessories/1898-mechanical-buzzer-with-led-indicator-ad16-22sm-robotics-bangladesh.html>

4.7 Power System

- Components
 - 18650 battery
 - TP4056 charging module
- Features
 - Rechargeable
 - Portable
 - Price: 100
 - Reference: <https://store.roboticsbd.com/battery-charger/1687-37v-18650-rechargeable-battery-solderable-robotics-bangladesh.html>

Alternative

- Solar system
 - Price: 500 to 700 BDT
 - Best for outdoor

4.8 Supporting Components

- Breadboard
- Jumper wires
- Waterproof box
- Price: Varies

5. Total Cost Estimation

- Basic system cost: 4800 to 5500 BDT
- Cost varies depending on quality and supplier

6. Communication System

- Uses ESP32 built-in WiFi
- No extra module needed

Alternative communication

- GSM SIM800L
 - Price: 350 to 450 BDT
- LoRa module
 - Price: 600 to 900 BDT

7. IoT Platforms

- Blynk
 - Easy mobile interface
- ThingSpeak
 - Data visualization
- Firebase
 - Custom app development

8. Parameters Monitored

- Water temperature

- pH level
- Turbidity
- Device status

9. Smart Features and Improvements

Automation

- Auto aerator control
- Auto water pump control

Smart Monitoring

- Mobile notifications
- Real-time dashboard

Advanced Features

- Smart fish feeder
- Solar powered system
- SMS alert system
- Multi pond monitoring
- AI based prediction system

10. Advantages

- Low cost and affordable
- Real world application in Bangladesh
- Easy to build and expand
- Reduces manual monitoring
- Scalable to commercial level

11. Conclusion

- Provides efficient solution for fish farming problems
- Combines IoT and embedded systems effectively
- Suitable for academic project and real life use
- Can evolve into startup level product with upgrades